Printed Pages : 3 **Subject Code : EEC404**

Paper Id **131225** **Roll No.** | | | | | | | | | | |

BTECH
(SEM-IV) THEORY EXAMINATION 2018-19
SIGNALS AND SYSTEMS

Time : 3 Hours ***Max. Marks : 100***

Note : *Be precise in your answer. In case of numerical problem assume data wherever not provided*

SECTION – A

1. Attempt any five parts of the following questions: 5×4=20

(a) Explain Time variance and static properties of a system.

(b) Write the condition for existence of DTFT.

(c) What is relationship between Fourier Transform and Laplace Transform?

(d) Find even and odd part of signal x (t) = e^{-2t}u (t).

(e) Determine whether or not signal x (t) = $\cos\frac{2\pi n}{7}\ \sin\frac{3\pi n}{14}$ is periodic .if periodic find its fundamental period.

(f) Determine whether or not signal x (t) = cos (t+1) x (t) is causal and time variant.

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SECTION B

2. Attempt any three parts of the following questions: 3×10=30

(a) When the input to an LTI system is $x(n) = (\frac{1}{3})^n u(n) + 2^n u(-n+1)$ and the corresponding $y(n) = 5.(\frac{1}{5})^n\ u(n) - 5.(\frac{2}{3})^n\ u(n)$.

(i) Find the function H(z) of the system and its ROC.

(ii) Find the impulse response h(n) of the system.

(b) Find the continuous time Fourier transform of the Gate/Rectangular signal. Also plot its magnitude response.

(c) Given $x(t) = 5\ cos\ t, y(t) = 2\ e^{-t}$, find the convolution of x(t) and y(t) using Fourier transform.

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(d) A signal x (t) has the Laplace Transform X(s) = $\frac{(s+2)}{(s^2+4s+5)}$. Find the Laplace Transform of the following signal y (t) = x (2t - 1) u (2t -1).

(e) Determine the Z-Transform of following sequences with ROC

(i) x(n) = $\left(\frac{1}{2}\right)^n$; $0 \leq n \leq 6$

x(n) = 0 otherwise

(ii) x(n) = n^2u(n) + nu(n)

SECTION C

3. Attempt any one parts of the following : 1×10=10

(a) (i) Determine the step response of the system described by difference equation y (n) + 4y(n-1) + 4y(n-2) = x(n).

(ii) Find the response of the time invariant system with impulse response h(n) = (1 , 2 , 1 , -1) to an input signal x(n) = (1 , 2 , 3 , 1) by using Z-Transform.

(b) Suppose that x(t) = $e^{-(t-2)}$ u(t-2) and h(t) is shown in Figure 1. Verify the convolution property for this pair.

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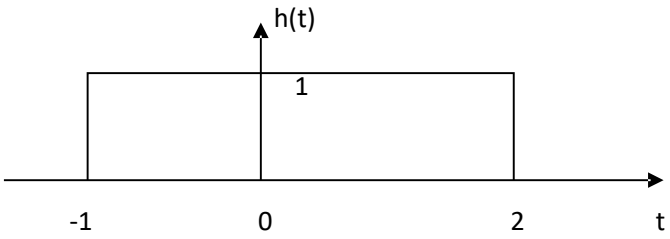


Figure 1

4. Attempt any one parts of the following : 1×10=10

(a) If $X(s) = \frac{2s+3}{(s+1)(s+2)}$ find x(t) for

i). System is stable

ii). System is causal

iii). System is non causal

(b) For the discrete time system $y(n) = x(-n+2)$ and $y(n) = x(n) + n.x(n+1)$, check whether the system is static or dynamic, linear or non-linear, shift invariant or shift varying, causal or non-causal.

5. Attempt any one parts of the following : 1×10=10

(a) Determine the total response of the differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = x(t)$$

Where y(0)=3, y'(0)=4, $x(t)=4e^{-2t}$ and $t \geq 0$.

(b) Compute the Fourier transform of the following signals:

(1) $x(n) = 2^n u(-n)$

(2) $x(n) = (\frac{1}{4})^n u(n+4)$.

6. Attempt any one parts of the following : 1×10=10

(a) Determine the inverse Z-transform using partial fraction method for

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$$X(z) = \frac{(\frac{1}{4})z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right)}$$

(1) $|z| > \frac{1}{2}$ (2) $|z| < \frac{1}{4}$ (3) $\frac{1}{4} < |z| < \frac{1}{2}$

(b) Given the relationship $y(t) = x(t) * h(t)$ and $g(t) = x(3t) * h(3t)$, and given that X(t) has Fourier transform X(jω) and h(t) has Fourier transform H(jω). Use Fourier transform properties to show that g(t) has the form $g(t) = Ay(Bt)$. Determine the value of A & B.

7. Attempt any one parts of the following : 1×10=10

(a) Compute the DTFT of $x(n) = (b^n \sin \omega_0 n)\, u(n); b < 1$.

(b) (i) Determine output response of system h(n) = (1,0,1,0) for input x(n) = (1,2,3,4).

(ii) Give and explain the relation between S-plane and Z-plane.

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